

BREAST CANCER CLASSIFICATION USING ARTIFICIAL NEURAL NETWORK (ANN)

Full Professional Project Documentation

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Final Test Accuracy: 98.25%

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1. ABSTRACT

Breast cancer is an important health problem worldwide, and early identification can support timely medical decision-making. This project develops an Artificial Neural Network (ANN) for classifying breast tumors as Benign or Malignant using numerical medical features.

The dataset was preprocessed by removing the identification column, encoding target labels, splitting the data into training and testing sets, and applying feature scaling. A multi-layer ANN was trained using the Adam optimizer and Binary Crossentropy loss function.

The final model achieved a Test Accuracy of 98.25% with a Test Loss of approximately 0.1527. These results demonstrate excellent performance on the unseen test data. This project is intended for educational and research purposes and is not a replacement for professional medical diagnosis.

2. INTRODUCTION

Artificial Intelligence and Machine Learning are increasingly used to analyze complex datasets and discover useful patterns. In healthcare-related research, machine-learning models can support prediction and classification tasks.

This project focuses on classifying breast tumor cases into Benign (non-cancerous) and Malignant (cancerous). An Artificial Neural Network was selected because neural networks can learn non-linear relationships among multiple input features.

3. PROBLEM STATEMENT

The problem addressed in this project is to build and evaluate an Artificial Neural Network capable of accurately classifying breast tumor cases as Benign or Malignant based on the medical features available in the dataset.

4. PROJECT OBJECTIVES

- Load and understand the Breast Cancer dataset.
- Explore dataset structure, columns and target classes.
- Perform data preprocessing and remove unnecessary information.
- Encode categorical target values into numerical values.
- Split the data into training and testing sets.
- Scale numerical features using StandardScaler.
- Build and train an Artificial Neural Network.
- Evaluate the model using test accuracy and test loss.
- Generate predictions for unseen test samples.
- Analyze results using a Classification Report and Confusion Matrix.

5. DATASET DESCRIPTION

Dataset Name: Breast Cancer Dataset

Dataset Path: /kaggle/input/datasets/yasserh/breast-cancer-dataset/breast-cancer.csv

Total Records: 569

Original Columns: 32

Target Column: diagnosis

Problem Type: Binary Classification

6. DATASET FEATURES

The dataset contains numerical measurements describing characteristics of breast tumors. Important examples include radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry and

fractal dimension. Many measurements are represented as mean, standard error and worst values.

Examples of features include radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, compactness_mean, concavity_mean, symmetry_mean and fractal_dimension_mean.

7. TARGET VARIABLE

The target column is diagnosis. It contains two classes:

B = Benign = encoded as 0

M = Malignant = encoded as 1

The labels were converted into numerical values using LabelEncoder for ANN training.

8. DATA PREPROCESSING

The dataset was loaded with Pandas. The id column was removed because it is an identifier rather than a useful predictive medical feature. Input features and the target variable were separated as X and y. The diagnosis labels were converted into numerical values using LabelEncoder.

Main preprocessing steps:

- Load dataset
- Check shape, columns and missing values
- Remove id column
- Separate X and y
- Encode diagnosis labels

9. TRAIN-TEST SPLIT

The dataset was divided into training and testing portions using an 80/20 split with stratification.

Training Samples: 455

Testing Samples: 114

Total Records: 569

10. FEATURE SCALING

StandardScaler was used to standardize numerical features. The scaler was fitted only on the training data and then applied to the test data to avoid data leakage.

Feature scaling helps ANN training by placing features on comparable numerical scales and improving convergence.

11. ARTIFICIAL NEURAL NETWORK (ANN)

An Artificial Neural Network is composed of interconnected layers of neurons. During training, the network adjusts its weights to reduce prediction error and learn relationships between input features and the target class.

For this binary classification problem, the network receives scaled tumor features and predicts the probability of the Malignant class.

12. ANN MODEL ARCHITECTURE

Input Layer: 30 features

Dense Layer 1: 64 neurons with ReLU activation

Dense Layer 2: 32 neurons with ReLU activation

Dropout Layer: 20%
Dense Layer 3: 16 neurons with ReLU activation
Output Layer: 1 neuron with Sigmoid activation

The model contained 4,609 trainable parameters.

13. ACTIVATION FUNCTIONS

ReLU (Rectified Linear Unit) was used in the hidden layers to introduce non-linearity and support learning of complex patterns.

Sigmoid was used in the output layer because this is a binary classification problem. The output represents a probability between 0 and 1.

14. DROPOUT LAYER

A Dropout layer with a rate of 0.20 was included. During training, Dropout randomly disables a proportion of neurons, which can help reduce overfitting and improve generalization.

15. MODEL COMPILATION

Optimizer: Adam
Loss Function: Binary Crossentropy
Evaluation Metric: Accuracy

16. MODEL TRAINING

Epochs: 50
Batch Size: 16
Validation Split: 20% of the training data

The model was trained for 50 epochs while validation performance was monitored during training.

17. TRAINING RESULTS

Validation Accuracy: 95.60%
Validation Loss: approximately 0.137

The training results show strong learning. Training accuracy became very high while validation accuracy remained around the mid-95% range, so generalization should continue to be monitored.

18. MODEL EVALUATION

After training, the ANN was evaluated using the unseen test dataset containing 114 samples.

Test Loss: 0.1526673287153244
Test Accuracy: 98.25%

The final test accuracy indicates excellent performance on the selected unseen test split.

19. CONFUSION MATRIX

The confusion matrix result was:

Actual Benign predicted as Benign: 72
Actual Benign predicted as Malignant: 0
Actual Malignant predicted as Benign: 2

Actual Malignant predicted as Malignant: 40

Therefore, 112 predictions were correct and 2 predictions were incorrect.

20. CLASSIFICATION METRICS

A Classification Report provides Precision, Recall and F1-score for each class.

Precision measures the correctness of predictions for a class. Recall measures how many actual cases of a class were successfully identified. The F1-score combines precision and recall into one useful measure.

21. ACCURACY CALCULATION

Correct Predictions = 72 + 40 = 112

Total Test Samples = 114

Accuracy = $(112 / 114) \times 100 = 98.25\%$

22. PERFORMANCE ANALYSIS

Strengths:

- Excellent final test accuracy of 98.25%.
- Only 2 incorrect predictions among 114 test samples.
- No Benign test case was classified as Malignant in the confusion matrix.
- The model learned effectively from standardized numerical features.
- Dropout was included to support generalization.

Limitations:

- The dataset contains only 569 records and is relatively small.
- Very high training accuracy means generalization should be monitored with additional validation methods.
- Accuracy alone does not describe every aspect of class-specific performance.
- A medical classification model requires extensive validation before real-world clinical use.

23. TECHNOLOGIES AND LIBRARIES USED

Python: Programming language

Pandas: Data loading and manipulation

NumPy: Numerical operations

Matplotlib: Visualization

Seaborn: Confusion matrix visualization

Scikit-learn: Data splitting, scaling and evaluation

TensorFlow: Deep learning framework

Keras: ANN model construction and training

Kaggle: Notebook and dataset environment

24. PROJECT WORKFLOW

1. Load Breast Cancer Dataset
2. Inspect Dataset Structure
3. Check Columns and Missing Values
4. Remove the ID Column
5. Separate Features (X) and Target (y)
6. Encode Diagnosis Labels
7. Split Data into Training and Testing Sets
8. Apply StandardScaler

9. Build ANN Architecture
10. Compile the Model
11. Train for 50 Epochs
12. Evaluate on Test Data
13. Generate Predictions
14. Create Classification Report
15. Create Confusion Matrix
16. Analyze Final Performance

25. FINAL PROJECT RESULTS

Total Records: 569

Training Samples: 455

Testing Samples: 114

Model: Artificial Neural Network (ANN)

Validation Accuracy: 95.60%

Validation Loss: approximately 0.137

Test Loss: approximately 0.1527

Final Test Accuracy: 98.25%

Correct Test Predictions: 112 out of 114

Overall Performance: EXCELLENT

26. CONCLUSION

This project successfully developed an Artificial Neural Network for breast cancer classification. The workflow included data loading, preprocessing, removal of the identifier column, target encoding, train-test splitting and feature scaling.

The ANN used multiple Dense layers, ReLU activations, a Dropout layer and a Sigmoid output layer. The final model achieved a Test Accuracy of 98.25% with a Test Loss of approximately 0.1527. The confusion matrix showed 112 correct predictions out of 114 test samples.

Overall, the experiment demonstrated excellent performance on the selected test set. The model should be viewed as an educational and research implementation rather than a replacement for professional medical diagnosis.

27. FUTURE WORK

- Use EarlyStopping to stop training when validation performance stops improving.
- Perform hyperparameter tuning for neurons, learning rate, batch size and epochs.
- Use additional validation strategies such as cross-validation.
- Compare ANN with Logistic Regression, SVM, Random Forest and XGBoost.
- Perform feature selection and additional feature analysis.
- Test alternative ANN architectures and regularization methods.
- Develop a web-based interface for educational demonstration.
- Evaluate the model on additional independent datasets.

END OF PROJECT DOCUMENTATION